

# Music Genre Recognition via Improved CRNN Architecture Using ResNet, BiLSTM, and Attention

Ahmet Selim Pala, Muhammed Ebrar Torun, Mesut Anlak, Ismail Emin Turhan

MYZ307E – Machine Learning for Electrical and Electronics Engineering

Istanbul Technical University

GitHub Repository: <https://github.com/mrebrartorun/music-genre-recognition-crnn>

**Abstract**—Music genre recognition is a representative audio signal classification problem that requires extracting meaningful spectral and temporal patterns from music recordings. In this project, a legacy Keras-based Convolutional Recurrent Neural Network (CRNN) architecture was reproduced in PyTorch and adapted from multi-label music tagging to 10-class single-label genre classification on the GTZAN dataset. After preprocessing and cleaning, 929 valid tracks were represented as segmented mel-spectrograms. The reproduced CRNN baseline achieved 76.84% validation accuracy and a macro F1-score of 0.760. To improve the baseline, a residual CNN front-end, a two-layer Bidirectional LSTM, and temporal attention pooling were introduced. The full improved model achieved 82.16% validation accuracy and a macro F1-score of 0.819, corresponding to a 5.32 percentage point improvement over the reproduced baseline. Ablation results show that the residual CNN front-end provides the dominant performance gain, while BiLSTM and attention contribute smaller incremental improvements.

## I. INTRODUCTION

Music Genre Recognition (MGR) aims to automatically assign music recordings to genre categories such as classical, jazz, metal, or rock. Although this task may appear simple from a user perspective, it is technically challenging because genre labels are often subjective, and different genres can share similar rhythmic, harmonic, and timbral characteristics. Therefore, hand-crafted rules are generally insufficient for robust classification.

Deep learning-based audio classification provides a data-driven approach to this problem by learning discriminative features directly from time-frequency representations. In this project, raw audio recordings are represented using mel-spectrogram segments, allowing convolutional layers to capture local spectral patterns and recurrent layers to model temporal dependencies.

The project follows a reproduce-and-extend methodology. First, the CRNN architecture proposed by Choi et al. [1] and its legacy Keras implementation [2] were analyzed and adapted to the GTZAN music genre classification task. Then, an improved architecture was designed by adding a residual CNN front-end, Bidirectional LSTM layers, and temporal attention pooling.

The main contributions of this work are fourfold: reproducing a legacy Keras CRNN architecture in PyTorch, adapting the original multi-label tagging model to GTZAN 10-class classification, improving the baseline with residual CNN, BiLSTM, and attention components, and evaluating the contribution of each component through ablation experiments.

## II. METHODOLOGY

### A. Reference Baseline

The reference work is based on the Convolutional Recurrent Neural Network proposed for music classification by Choi et al. [1]. The original GitHub implementation was designed for multi-label music auto-tagging and produced 50 tag probabilities using a sigmoid output layer [2]. Since our target task is GTZAN-based single-label classification, the model was adapted to output 10 genre classes.

The reproduced baseline consists of four convolutional blocks followed by two GRU layers and a fully connected classifier. The original Keras code was reimplemented in PyTorch to ensure compatibility with a modern training environment and to provide a flexible baseline for further improvements.

### B. Dataset and Preprocessing

The GTZAN dataset contains 10 music genres: blues, classical, country, disco, hiphop, jazz, metal, pop, reggae, and rock [3]. GTZAN is a widely used benchmark dataset, but previous studies have reported duplicated, mislabeled, or otherwise problematic files [4]. Therefore, the dataset was cleaned before training.

After cleaning and preprocessing, 929 valid audio tracks were used. Each track was represented as 40 mel-spectrogram segments of size  $128 \times 130$ , resulting in 37,160 segment-level samples. The data were split at the song level into 743 training files and 186 validation files, corresponding to 29,720 training segments and 7,440 validation segments. Splitting at the song level was preferred to reduce the risk of having segments from the same song in both training and validation sets.

### C. Improved Architecture

The improved model extends the baseline CRNN with three main architectural changes. First, the standard convolutional front-end was replaced with a residual CNN front-end to improve spectral feature extraction and reduce information loss through deeper convolutional processing. Second, the recurrent part was upgraded to a two-layer Bidirectional LSTM to capture temporal dependencies in both forward and backward directions. Third, temporal attention pooling was added to allow the model to assign higher importance to more discriminative time frames.

The final architecture can be summarized as:

Mel-Spectrogram → Residual CNN  
→ BiLSTM → Attention  
→ Classifier.

Cross-entropy loss was used for single-label classification, and the Adam optimizer was used during training. The same song-level train-validation split was used for all experiments to ensure a fair comparison between the reproduced baseline and the improved variants. Model performance was evaluated using validation accuracy, macro F1-score, training history, and confusion matrix analysis.

In addition to the full model, two ablation variants were evaluated: a ResNet-only model and a ResNet+BiLSTM model without attention. This allowed us to measure the contribution of each architectural component separately.

### III. RESULTS

#### A. Overall Performance

Table I summarizes the validation accuracy, macro F1-score, best epoch, and improvement over the reproduced baseline.

TABLE I  
COMPARISON OF BASELINE, ABLATION MODELS, AND FULL IMPROVED MODEL

| Model              | Acc.          | F1           | Epoch | $\Delta$        |
|--------------------|---------------|--------------|-------|-----------------|
| Baseline CRNN      | 76.84%        | 0.760        | 9     | –               |
| ResNet-only        | 81.72%        | 0.816        | 13    | +4.88 pp        |
| ResNet+BiLSTM      | 81.84%        | 0.816        | 12    | +5.00 pp        |
| Full Improved CRNN | <b>82.16%</b> | <b>0.819</b> | 15    | <b>+5.32 pp</b> |

The reproduced PyTorch CRNN baseline achieved 76.84% validation accuracy. The full improved model achieved 82.16%, corresponding to a 5.32 percentage point improvement. This indicates that the proposed modifications improved the model’s ability to extract and combine spectral-temporal information from music segments.

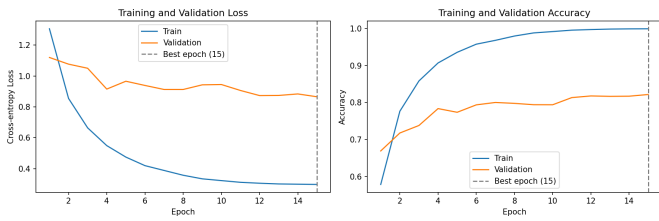


Fig. 1. Training and validation curves of the full improved CRNN model.

#### B. Ablation Study

The ablation study shows that the residual CNN front-end is the most influential modification. The ResNet-only model increased validation accuracy from 76.84% to 81.72%, accounting for most of the total performance gain. Adding BiLSTM without attention increased accuracy only slightly to 81.84%. The full model with attention reached 82.16%,

showing that attention provided an additional but smaller improvement.

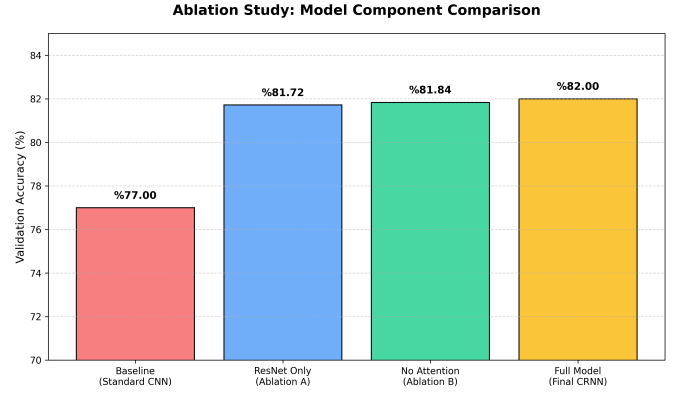


Fig. 2. Ablation study comparison of the reproduced baseline, ResNet-only variant, ResNet+BiLSTM without attention, and full improved model.

This result suggests that stronger convolutional feature extraction is the dominant factor in this experiment. BiLSTM and attention still improve performance, but their contribution is more incremental compared with the residual CNN front-end. This finding is useful because it shows that adding recurrent and attention mechanisms does not automatically guarantee large improvements unless the feature extractor is already strong enough.

#### C. Class-Level Analysis

The improved model performed especially well on genres with distinctive spectral or rhythmic structures, such as classical and metal. However, some classes remained difficult. Rock, country, blues, and reggae showed stronger confusion patterns, which is expected because these genres may share similar instrumentation and rhythmic characteristics.

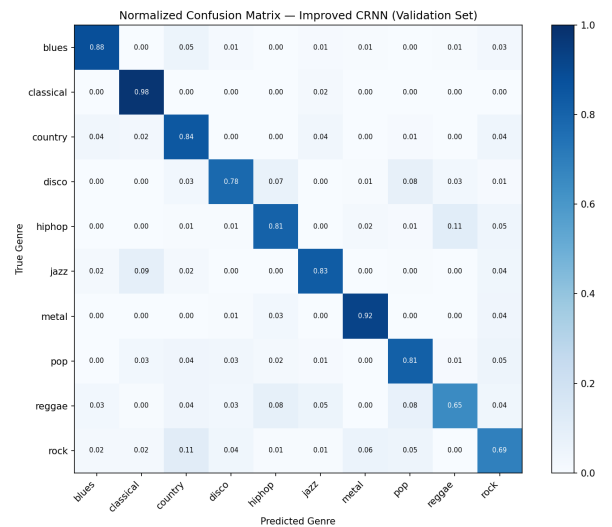


Fig. 3. Confusion matrix of the full improved CRNN model on the validation set.

The confusion matrix indicates that the model can separate several genres reliably, but acoustic similarity between certain genre pairs still limits performance. This suggests that future improvements should focus not only on model complexity, but also on more robust preprocessing, augmentation, and song-level aggregation strategies.

#### IV. CONCLUSION

This project reproduced a legacy Keras-based CRNN music classification architecture in PyTorch and adapted it to GTZAN 10-class music genre recognition. The reproduced baseline achieved 76.84% validation accuracy. By introducing a residual CNN front-end, BiLSTM layers, and temporal attention pooling, the full improved model reached 82.16% validation accuracy and a macro F1-score of 0.819.

The ablation study demonstrated that the residual CNN front-end was responsible for the majority of the improvement, while BiLSTM and attention provided smaller additional gains. These results show that improving the convolutional feature extractor is particularly important for mel-spectrogram-based music genre classification.

The results also indicate that increasing architectural complexity alone does not always lead to proportional gains. In this study, most of the improvement was obtained from the residual convolutional feature extractor, while the recurrent and attention components mainly refined the final decision. This finding is important for future work, since it suggests that feature extraction quality is a key factor for mel-spectrogram-based music genre recognition.

Future work may include song-level majority voting instead of purely segment-level evaluation, more systematic hyperparameter tuning, stronger augmentation strategies, and comparison with transfer learning-based audio models. In addition, further analysis of confused genres such as rock, country, blues, and reggae may help identify more targeted improvements.

#### REFERENCES

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